

i.a) create



pair splits

i.b) move



spinon walks

i.c) annihilate

 $\mathbf{q}_\gamma = (0, 0, 0)$

i.d) create



pair splits

no walk:
the loop closes
at second order

i.e) annihilate

 $\mathbf{q}_\gamma = (0, 0, -1)$ 